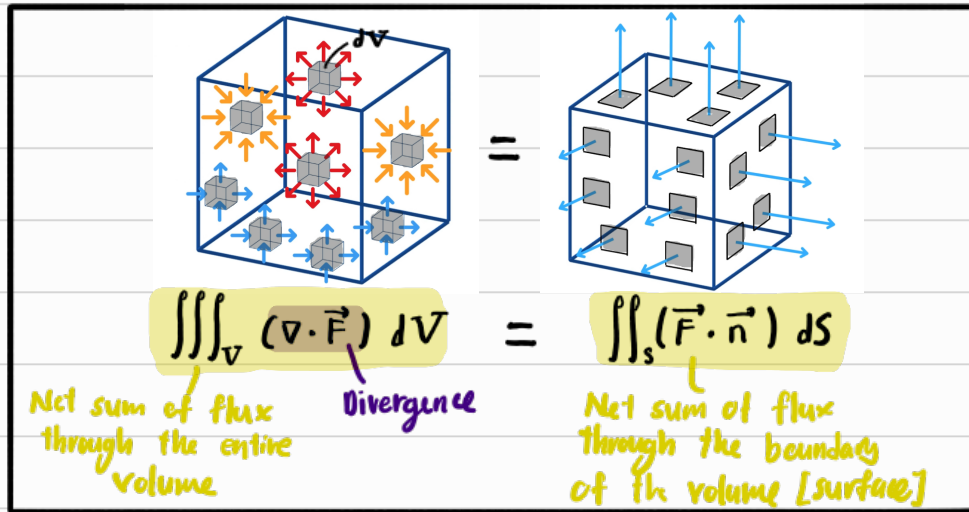


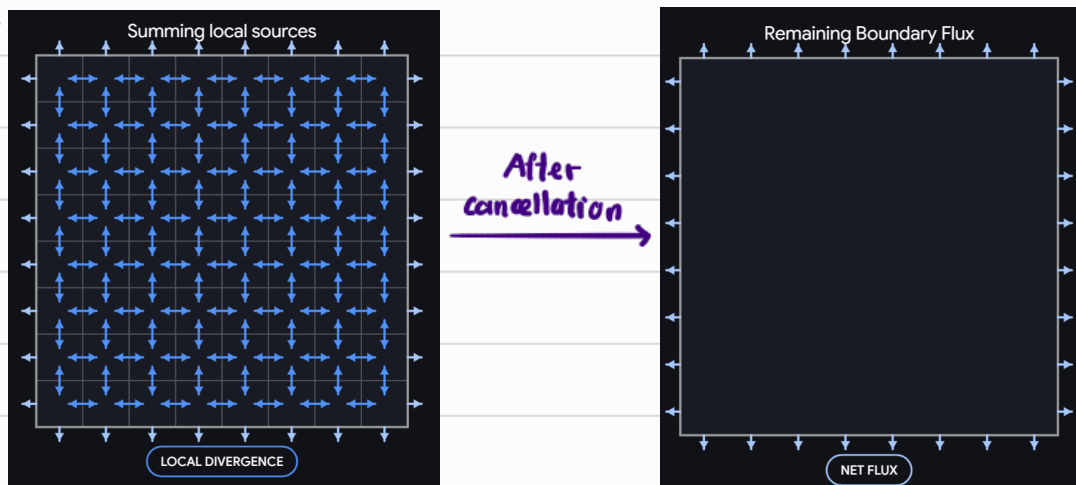
Gauss Divergence Theorem

Total ammount of "stuff" being created or destroyed inside a closed space is exactly equal to the net ammount of "stuff" flowing out or into its outer surface (shell)



Explanation

Each microscopic cube, dV shares a wall. Because they share a wall, the outward flux of one dV will be the inward flux of the neighbouring dV and vice versa.



$\therefore$ After cancelling the flux of every single shared internal wall, we are left w the flux at the boundary / outer surface of the volume.

Conditions for divergence theorem

The Divergence Theorem Let S be a ^①closed, ^②piecewise smooth, orientable surface which encloses a volume V . If $\mathbf{F}$ is a three dimensional vector field with components which have continuous first order partial derivatives ^③throughout V then ^④

$$\iint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) dS = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

where $\hat{\mathbf{n}}$ is an outward pointing unit normal to the surface S .

① Closed Surface

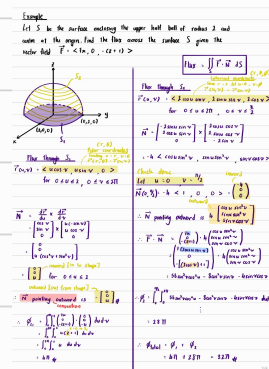
② Piecewise smooth ~ Made up of flat/smoothly curving surface

③ Orientable ~ Has a clearly defined inside/outside

④ Continuous 1st order partial ~ All partials are continuous [aka. no holes]

Example [same example in week 5]

Let S be the surface enclosing the upper half ball of radius 2 and centre at the origin. Find the flux across the surface S given $\vec{F} = \langle 7x, 0, -(z+1) \rangle$



$$\begin{aligned} \text{Flux across } S &= \iint_S \vec{F} \cdot \vec{n} dS \\ &= \iiint_V \nabla \cdot \vec{F} dV \end{aligned}$$

$$= \iiint_V \begin{pmatrix} \frac{d}{dx} \\ \frac{d}{dy} \\ \frac{d}{dz} \end{pmatrix} \cdot \begin{pmatrix} 7x \\ 0 \\ -(z+1) \end{pmatrix} dV$$

$$= \iiint_V 6 dV$$

$$= 6 \iiint_V 1 dV \quad \text{Volume of hemisphere}$$

$$= 6 \left[\frac{4}{3} (2^3) / 2 \right]$$

$$= 32\pi$$

Stokes Theorem

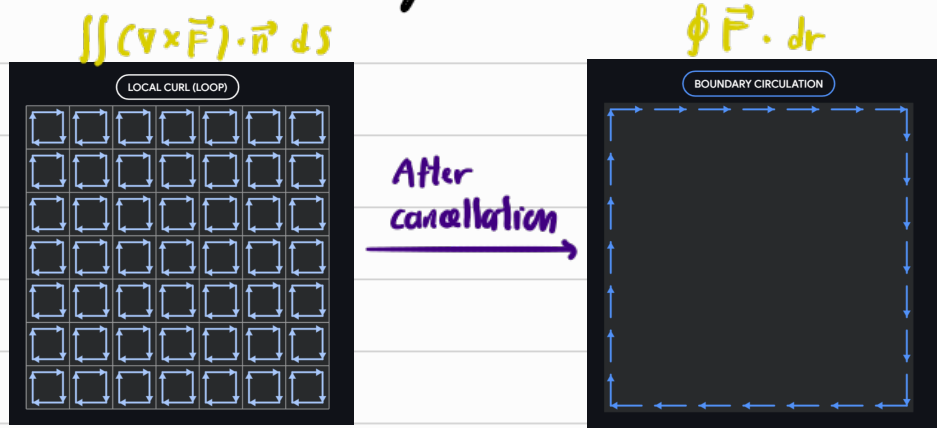
This is just Gauss divergence Theorem but instead of divergence, its curl.
And instead of comparing Volume & surface, its surface & line.

$$\iint (\nabla \times \vec{F}) \cdot \vec{n} \, dS = \oint \vec{F} \cdot d\vec{r}$$

Net sum of curl over the surface Net sum of curl along the boundary of the surface [Line]

Explanation

Same explanation as divergence theorem.



Conditions for Stokes theorem

Stokes' Theorem

Let S be an oriented surface in $\mathbb{R}^3$, with normal $\hat{n}$, bounded by a positively oriented curve C . If $\vec{F}$ is a three dimensional vector field with components which have continuous first order partial derivatives throughout S then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

① Oriented ~ Distinct inside / outside

② Bounded by positive oriented curve ~ Right hand rule [Thumb = $\hat{n}$, finger curl = direc]

③ Continuous 1st order partial ~ All partials are continuous [aka. no holes]

Green's Theorem

Stokes Theorem but only in the xy plane.

$$\oint_C \vec{F} \cdot d\vec{r}$$

$$\iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, dS$$

$$\oint_C \left(P(x,y) \frac{dx}{dt} + Q(x,y) \frac{dy}{dt} \right) dt = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Explanation:

$$\text{Stokes Theorem: } \oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, dS$$

$$\text{Since we are in the xy plane} \Rightarrow \vec{F}(x,y) = \langle P(x,y), Q(x,y), 0 \rangle$$

$$\Rightarrow d\vec{r} = \langle dx, dy, 0 \rangle$$

$$\Rightarrow \vec{n} = \langle 0, 0, 1 \rangle$$

$$\therefore \vec{F} \cdot d\vec{r} = \begin{pmatrix} P \\ Q \\ 0 \end{pmatrix} \cdot \begin{pmatrix} dx \\ dy \\ 0 \end{pmatrix} = P(x,y) dx + Q(x,y) dy$$

$$\Rightarrow \oint_C P dx + Q dy = \oint_C P \frac{dx}{dt} + Q \frac{dy}{dt} dt$$

$$\therefore \nabla \times \vec{F} = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \times \begin{pmatrix} P \\ Q \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \end{pmatrix}$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = \langle 0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \rangle \cdot \langle 0, 0, 1 \rangle = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

$$\therefore \iint (\nabla \times \vec{F}) \cdot \vec{n} \, dS = \iint \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, dA$$

curved area flat area

